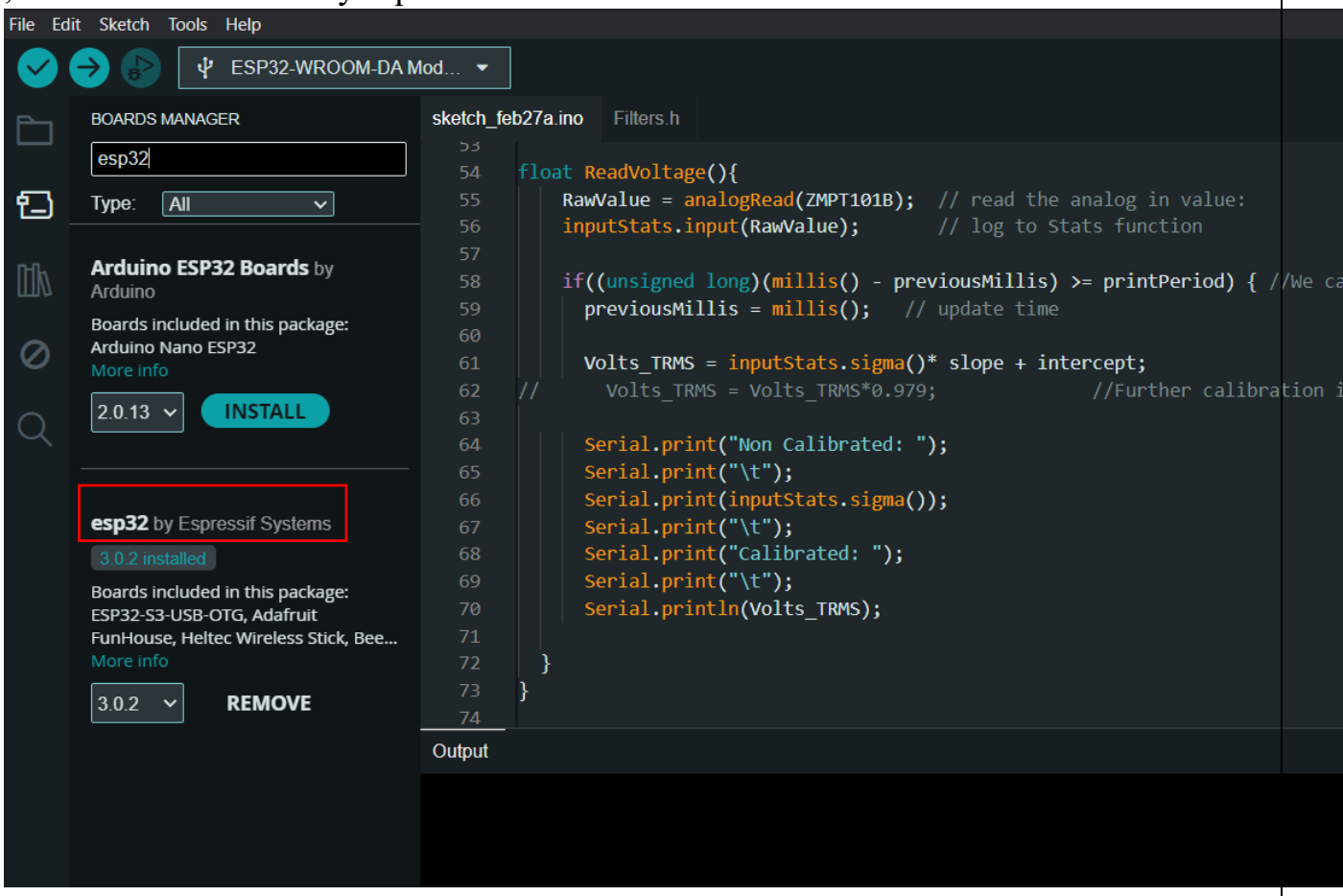
INSTRUCTION MANUAL FOR OVER THE AIR (OTA) FIRMWARE UPDATE FOR ESP32

Requirements

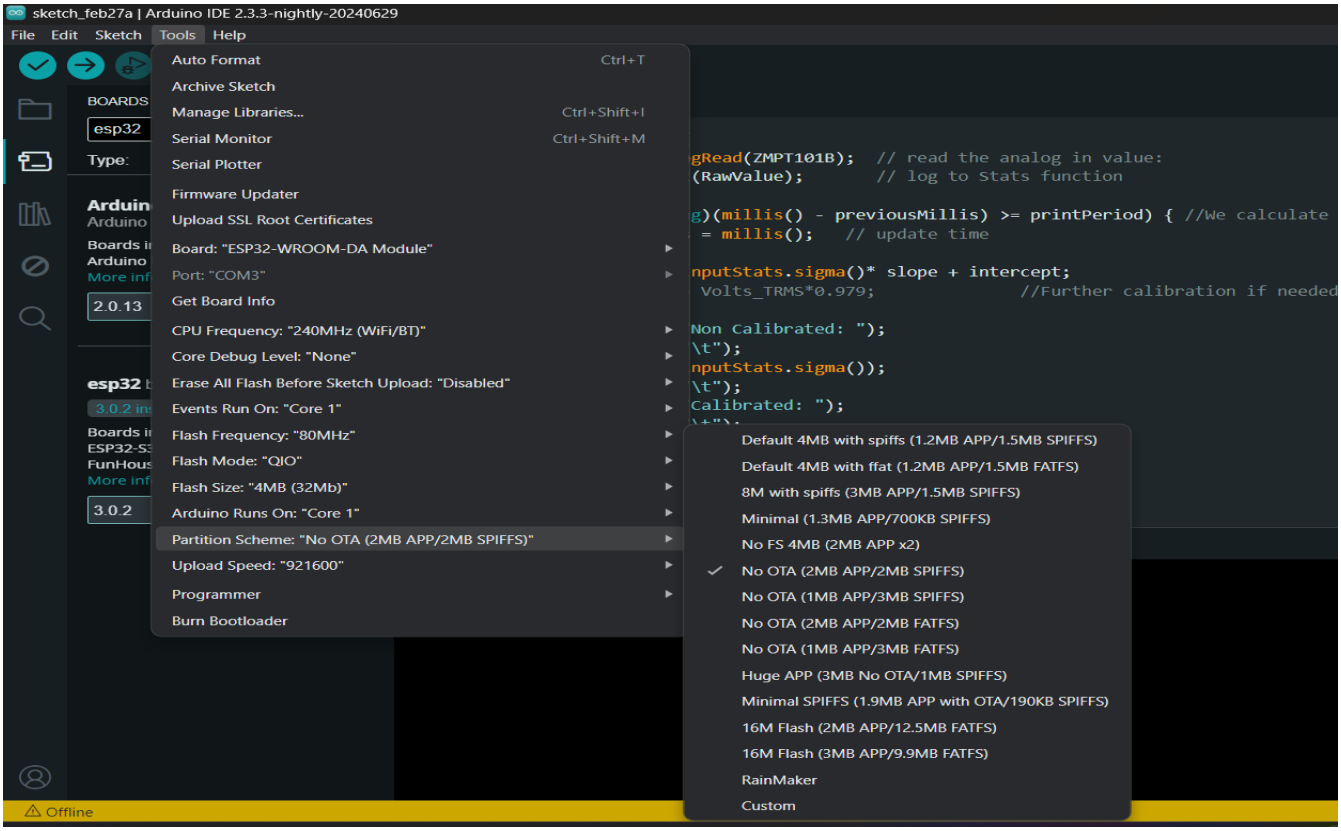
- ESP32
- Arduino IDE(latest version)
- Cable that supports data transfer to connect ESP32

Steps to follow

- Open Arduino IDE and create a new sketch
- In Boards manager select ESP32 WROOM DA MODULE. If it does not appear ,install ESP32 boards by espressif



- Go to tools menu and select the below configuration



- After configuring, write your code and save it.

POINTS to note while writing code

1. Interval is the time set for blinking of led
2. Update interval is time set for checking updates for firmware update
3. Both the values can be adjusted as per requirements

```
unsigned long previousMillis = 0;
const long interval = 1000;                    blinking the LED
bool ledState = false;

unsigned long updateCheckMillis = 0;
const long updateInterval = 60000;             hecking updates (1 hour)
```

4.

- Compile the program. Make sure to maintain the firmware version code in the program.

```
const int ledPin = 2; // GPIO pin for the LED

const char* firmwareUrl = "https://drive.usercontent.google.com/download?i

//https://drive.google.com/file/d/1RcWouLllIBlUguQVWMnSCk-y9mxzfIY8/view?u

#define CURRENT_FIRMWARE_VERSION "1.0.2"  this line with the new

unsigned long previousMillis = 0;
const long interval = 1000; // Interval for blinking the LED
bool ledState = false;

unsigned long updateCheckMillis = 0;
const long updateInterval = 60000; // Interval for checking updates (1 hou
```

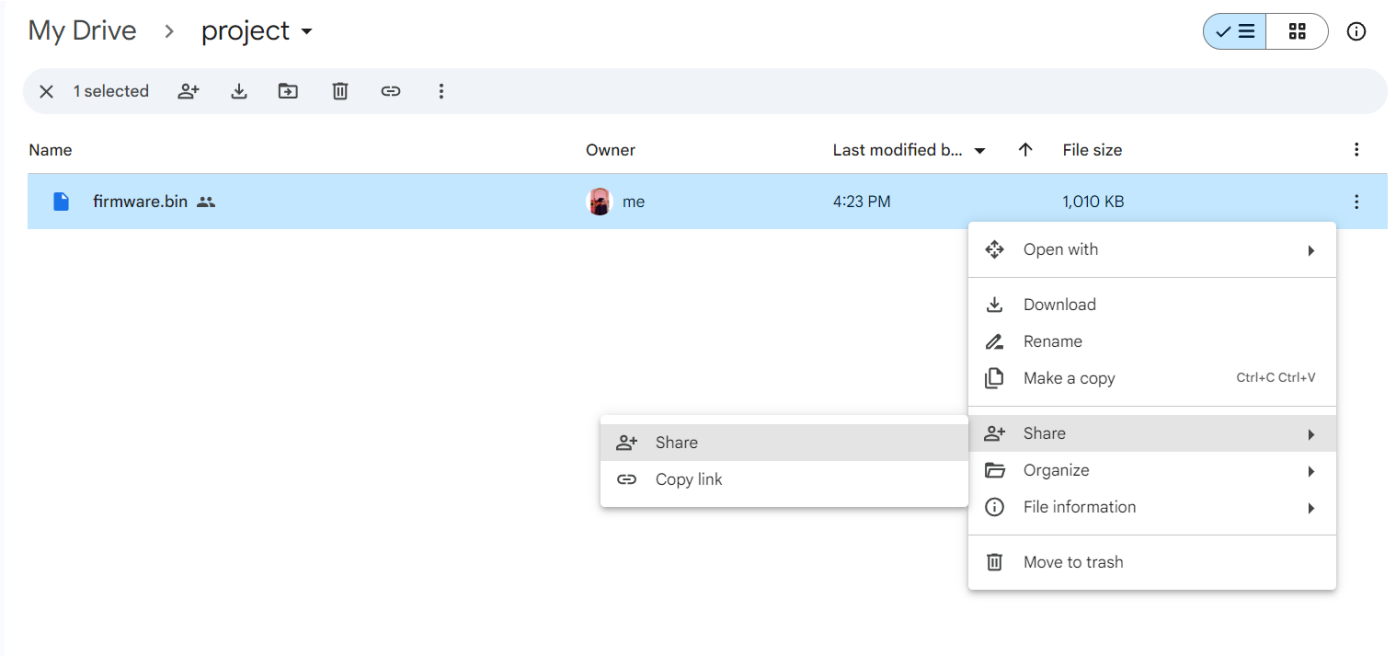
- Write the next version of the code in a separate new sketch and make sure to change the current firmware version as shown above.
- You should enter firmware version of the previous sketch in following variable

```
  if (canBegin) {
    Serial.println("Begin OTA update");
    WiFiClient* client = http.getStreamPtr();
    size_t written = Update.writeStream(*client);

    if (written == contentLength) {
      Serial.println("Written : " + String(written) + " successfully");

      // Compare the embedded version in the new firmware
      const char* newFirmwareVersion = "1.0.1";  should be read from
      if (strcmp(CURRENT_FIRMWARE_VERSION, newFirmwareVersion) != 0) {
        if (Update.end()) {
          Serial.println("OTA update finished!");
          if (Update.isFinished()) {
            Serial.println("Update successfully completed. Rebooting.");
            ESP.restart();
          } else {
            Serial.println("Update not finished? Something went wrong.");
```

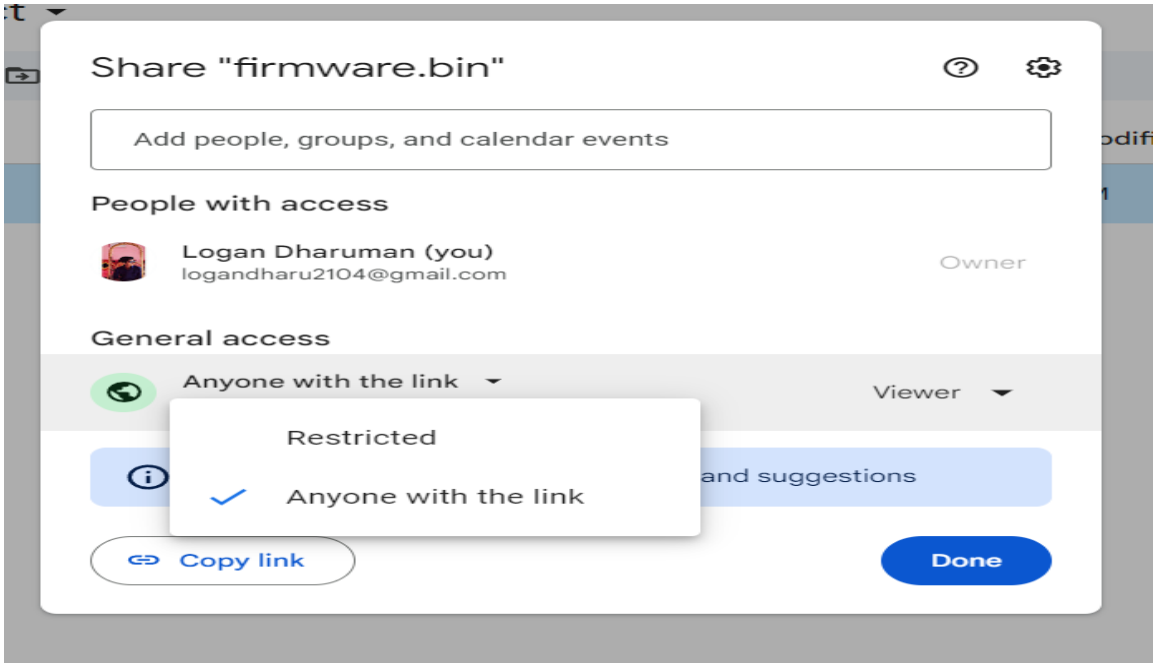
"1.0.1" is the current firmware version in the previous file.

- After writing the version 2 compile the program.
- After compiling the program go to sketch->export compiled binary.

- After compilation go to sketch->show sketch folder. You can find the BIN file for the version2 .
- Now open google drive and upload the created BIN file. Then go to more option and select share.



- Change access to anyone with the link and copy the link.



- Now change the link to the following format(direct download)

The shareable link typically looks like this:

https://drive.google.com/file/d/1RcWouLllIBlUguQVWMnSCk-y9mxzfIY8/view?usp=sharing

Copy the highlighted file id and paste it into the below highlighted place

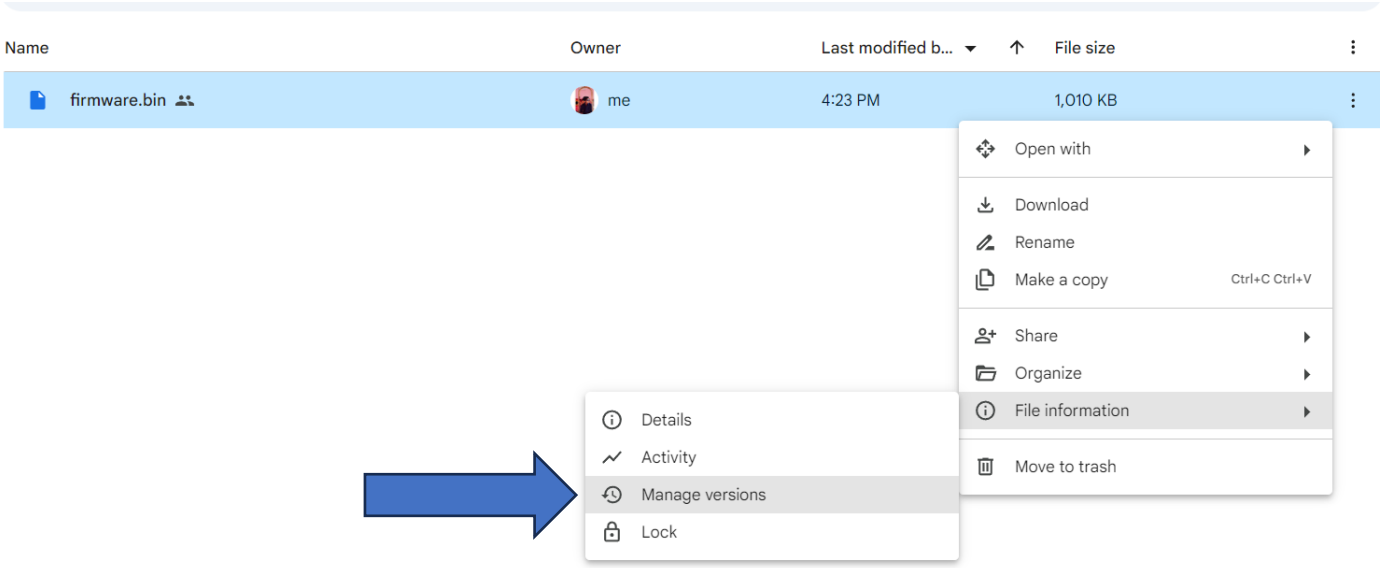
Now change it into

https://drive.usercontent.google.com/download?id=1RcWouLllIBlUguQVWMnSCk-y9mxzfIY8&export=download.

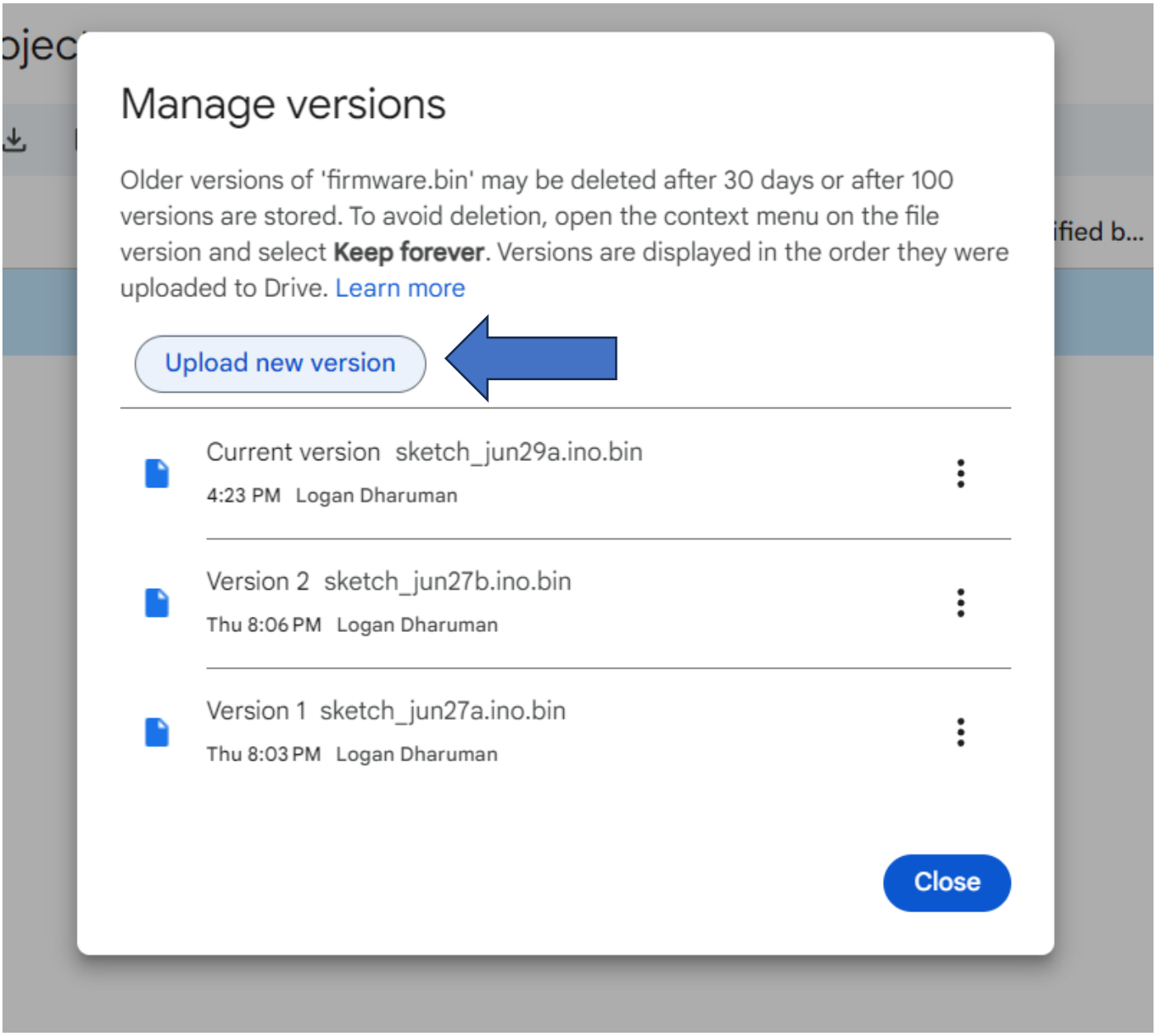
- Now place the direct download in the firmware url of the program in both verson 1 and version2

```
const char* ssid = "vivo 1938";
const char* password = "logan@2104";

const int ledPin = 2; // GPIO pin for the LED

const char* firmwareUrl = "https://drive.usercontent.google.com/download?id=1RcWouLllIBlUguQVWMnSCk-y9mxzfIY8&export=download";
```

- Now compile the version2 and create BIN file as in previous steps. After creating the BIN File , we are going to upload the updated bin file in the same link as in the following steps.



- Click manage versions and click upload new version.



- Now upload the updated BIN file from the sketch folder.
- Now compile and upload the version1 program to ESP32.
- You can now view the ongoing process in the serial monitor.
- After the specified update interval, the ESP32 checks for updates and checks for the firmware version of the file in the google drive. If the versions of the files are different ,the updated version is written in the esp32 and starts booting.
- Whenever you need to update the firmware of the esp32 , write a new program by changing the program version in the code and create a BIN file for the new code. After that, update the newly created bin file in the google drive by clicking manage versions and upload it.

ISSUES & HOW TO SOLVE IT:

1. When the update interval is reached and esp32 checks for update and it shows "Firmare not found at https://drive.usercontent.google.com/download?id=1RcWouLllIBlUguQVWMnSCk-y9mxzfIY8&export=download"

It is because the download link has become inaccessible.because of continuous requests to download the file.

It can be resolved by trying after sometime or by creating a google drive api key and including it to the program(complicated process).

Using google Drive to host the file is not recommended. It is recommended to have a dedicated web server to host the BIN file.

2. If you find the same version of the file installing again and again it is because you have not entered the previous firmware version in the following place

```
// Compare the embedded version in the new firmware
const char* newFirmwareVersion = "1.0.1"; // Th[illegible] be read
if (strcmp(CURRENT_FIRMWARE_VERSION, newFirmwareVersion) != 0) {
  if (Update.end()) {
    Serial.println("OTA update finished!");
    if (Update.isFinished()) {
      Serial.println("Update successfully completed. Rebooting.")
      ESP.restart();
```

3. Make sure to have stable internet connection or OTA updates may not begin.

Recommendations

Make sure you have installed the latest version of Arduino IDE.

Ensure all the required libraries are installed properly.

Highly recommended to use a web server to host the updated file.